Data Technician Week 5 Workbook Azure Data Fundamentals

Name:

Course Date:

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Day 1: Task 1

Please research and complete the below questions relating to key concepts of cloud.

Be prepared to discuss the below in the group following this task.

What can cloud computing do for us in the real-world?	Services like YouTube and Spotify use cloud to store/keep content and use it to stream it to many people
How can it benefit a business?	- Companies can save on electricity costs for cooling and space - Systems are updated regularly with the latest features which business never use exploitable tech - Allows business to deploy their applications near close to their global audience which means it increases the overall performance
What's the alternative to cloud computing?	On-Premises Infrastructure- owning and managing your own physical hardware locally Edge Computing

What cloud providers can we use, what are their features and functions?	AWS Pay-As-You-Go Pricing- Pay for specific cloud resources with no upfront costs or termination fees. Only pay for what you use Scalable Storage- automatically increase or decrease storage capacity in response to data demands without manual intervention Azure Advanced AI Machine Learning tools Hybrid and multi-cloud- run services across their own data centres and other clouds Offers Virtual Machines for running apps GCP Big Query for enterprise level data warehouses and Looker Studio for analytics AI Machine Learning tools- Vertex AI allowing use to use and manage AI models Global Infrastructure- built on the same infrastructure that supports Google's internal products (Gmail) which means it has a massive global network

Day 1: Task 2

Please research the below cloud offerings, explain what they are and examples of use cases.

Cloud Offerings	Explain what it is	When / how might you use this service in the real-world?
IaaS (Infrastructure as a service)	Cloud computing model that provides on-demand servers, storage and networking without the need for physical hardware	Business use it to process and analyse vast amounts of data efficiently by providing scalable storage Provides the necessary infrastructure for web apps. Businesses can quickly use web apps and scale resources up or down
PaaS (Platform as a service)	Cloud computing model that provides developers with a platform to build, deploy and manage applications	Businesses use PaaS to build custom analytics dashboards Banks use PaaS to build secure online portals for mobile banking and trading PaaS can be used to build virtual classrooms that can support many students simultaneously
SaaS (Software as a service)	Cloud based software delivery model where	Netflix provides streaming content via subscription and Google Drive offer cloud storage for personal files

	people access apps over the internet, while the cloud service provider handle security and updates	Microsoft Teams operates as a SaaS application within the Microsoft 365 suite, providing cloud-based collaboration tools without requiring local installation

Day 1: Task 3

Please research the below terms and explain what they are, when they would be appropriate and a real-world example of where it could be implemented (i.e. what type of organisation).

Public Cloud	Type of cloud computing where a third-party provider (such as AWS, Azure) delivers computing resources like apps to many users over the public internet Apps like Google Workspace uses public cloud infrastructure to allow teams to edit documents and message other people
Private Cloud	Type of cloud computing used exclusively by a single organisation Hospitals use private cloud to store electronic health records Banks use private cloud to handle personal financial data by making sure it doesn't interfere with other companies' data

Hybrid Cloud	Type of cloud computing that combines public and private cloud services, allowing transfer of data between them Hospitals store personal health details in a private cloud but use a public cloud for any research they want to conduct
Community Cloud	Type of cloud computing shared/used by multiple organisations that have similar needs Different government departments use community cloud to share resources between other agencies/organisations which improves teamwork Universities and schools use community cloud to share research information for similar projects that one might not have access to due to high costs/fees.

Day 2: Task 1

Describe, with examples, the **three** major areas that the Computer Misuse Act deals with.

Area	Description	Example
Unauthorised access to computer material	It occurs when someone logs onto a computer or network without permission granted	Using another person's logged in computer to read private email messages without their permission
Unauthorised access with intent to commit further offences	It occurs when someone hacks a computer with the intent to commit further crime like fraud or blackmail	An employee breaks into the company's payroll system with the intention of altering salary data to increase their own pay. This would fall under fraud
Unauthorised modification of computer material	This is the act of purposefully changing, deleting or damaging data Disrupting the overall performance of a computer through malware, viruses etc.	Creating and releasing a ransomware program that encrypts a hospitals' files, rendering them unusable until a ransom is paid

The computer misuse act 1990 is an act where an individual can be criminalised because of computer related offense. Describe three extra powers that the Police and Justice Act 2006 (Computer Misuse) has added.

Description

Criminalizing the supply/possession of hacking tools – You can't make, adapt, supply, obtain with intent "articles" for use in computer misuse offences
Addressing Denial of Service Attacks – The rule expanded to make it that intentional, unauthorised, disruptive actions are punishable offences in the UK; resulting in high- penalty criminal offences
Increased Penalties & Expanded Jurisdiction – The maximum sentence was increased from 5 to 10 years regarding unauthorised access with intent to commit further offences

Look at the below website to answer the questions:
https://www.gov.uk/personal-data-my-employer-can-keep-about-me

Write down three items of data which a company can store about an employee.
Name
Address
Date of Birth

Give three more examples of data that an employer can only store if they first get the employee's permission.

Religion
Genetics
Race and Ethnicity

Conduct further research to answer the below questions.

Question	Answer
Provide one example of: Copyright infringement	Recording a film while in a movie theatre
Provide one example of: Plagiarism	Copying text word for word from a website without citation
What are two consequences of copyright infringement and software piracy?	For severe and repeated offences, a person can face prison sentences up to 10 years Individuals can face lawsuits

Give three possible consequences for individuals when using pirated software	System crashes and lack of updates Heavy fines and prison sentences Malware virus

Listed below are some laws which we have covered today:

1. Computer Misuse Act 1990
2. Police and Justice Act 2006 (Computer Misuse)
3. Copyright, Designs and Patents Act 1988
4. Copyright (Computer Programs) Regulations 1992
5. The Health and Safety (Display Screen Equipment) Regulations 1992
6. Data Protection Act 2018
7. Consumer Rights Act 2015

- Insert a number in the first column of each row to match each of the statements with one of the above Acts.

- One of statements is incorrect and not illegal. For this statement, write 'Not illegal'.

Act number	Clause
3	With some exceptions, it is illegal to use unlicensed software

7	Any product, digital or otherwise, must be fit for the purpose it is supplied for
1	Unauthorised modification of computer material is illegal
Not illegal	It is illegal to create or use a hacking tool for penetration testing
6	Personal data may only be used for specified, explicit purposes
5	Employers must provide their computer users with adequate health and safety training for any workstation they work at
1	It is illegal to distribute hacking tools for criminal purposes
3	It is illegal to distribute an illicit recording
6	Personal data may not be kept longer than necessary
1	Gaining unauthorised access to a computer system is illegal
5	Employers must ensure that employees take regular and adequate breaks from looking at their screens
1	It is illegal to prevent or hinder access (e.g. by a denial-of-service attack) to any program or data held in any computer
6	Personal data must be accurate and where necessary kept up to date

Day 3: Task 1

Please complete the below lab (3) *'Explore relational data in Azure'* and paste evidence of the completed lab in the box provided.

Explore relational data in Azure

Learning Path 02 (CSS)

Duration:	2 Hours, 15 Minutes
Lab Series:	DP-900T00-A Microsoft Azure Data Fundamentals [Cloud Slice Provided]
Virtualization Platform:	Hyper-V
RAM:	6.5GB
Cloud Platform:	Azure
Content Version:	2
Is Exam:	No
Status:	Not Running

Launch

Completed lab

1

Explore Azure SQL Database (Expected Duration 2 hours, 15 minutes)

DP-900T00-A Introduction to Microsoft Azure Data [Cloud Slice Provided], Learning Path 02 (CSS)

Required: Yes

Started: 17 February 2026 18:41 (Central Standard Time)

Ended: 17 February 2026 19:06 (Central Standard Time)

Launch Cancel

9 of 10 launch attempts remaining

Day 3: Task 2

Please complete the below lab (4) *'Explore non-relational data in Azure'* and paste evidence of the completed lab in the box provided.

Explore non-relational data in Azure

Learning Path 03 (CSS)

Duration:	2 Hours, 15 Minutes
Lab Series:	DP-900T00-A Microsoft Azure Data Fundamentals [Cloud Slice Provided]
Virtualization Platform:	Hyper-V
RAM:	6.5GB
Cloud Platform:	Azure
Content Version:	2
Is Exam:	No
Status:	Not Running

Launch

Completed lab

2

Explore Azure Storage for non-relational data (Expected Duration 2 hours, 15 minutes)

DP-900T00-A Introduction to Microsoft Azure Data [Cloud Slice Provided], Learning Path 03 (CSS)

Required: Yes

Started: 17 February 2026 19:11 (Central Standard Time)

Ended: 17 February 2026 20:24 (Central Standard Time)

Launch Cancel

8 of 10 launch attempts remaining

Day 3: Task 3

Please complete the below lab (5) 'Explore data analytics in Azure' and paste evidence of the completed lab in the box provided.

Explore data analytics in Azure

Learning Path 04 (CSS)

Duration:	3 Hours
Lab Series:	DP-900T00-A Microsoft Azure Data Fundamentals [Cloud Slice Provided]
Virtualization Platform:	Hyper-V
RAM:	6.5GB
Cloud Platform:	Azure
Content Version:	2
Is Exam:	No
Status:	Not Running

Launch

Completed lab

3

Explore fundamentals of large-scale data analytics (Expected Duration 3 hours)

DP-900T00-A Introduction to Microsoft Azure Data [Cloud Slice Provided], Learning Path 04 (CSS)

Required: Yes

Started: 18 February 2026 17:50 (Central Standard Time)

Ended: 18 February 2026 19:45 (Central Standard Time)

Launch

8 of 10 launch attempts remaining

Day 4: Task 1

In your teams, complete the Azure DP-900 practice exam and paste your result below – this is open book and please research and discuss your answers as a team.

6

Practice Assessment: DP-900T00-A Microsoft Azure Data Fundamentals

Practice Assessment for Microsoft Certifications for DP-900T00-A

Additional Details

Required:	No
Available Instructor-Led:	Yes
Available Self-Paced:	Yes

Result

Overall Results

To be better prepared for the exam, aim to achieve a score of 80% or higher in multiple attempts.

Score: 60%

Show My Answers

Performance by assessment section

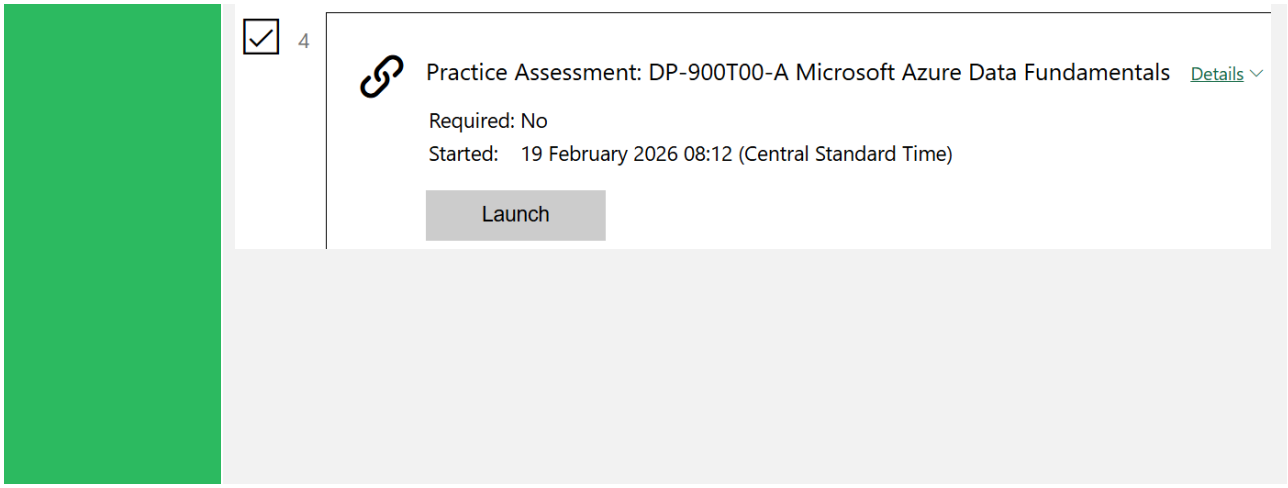
To further strengthen your skills in the following areas, refer to the Customized Learning Material section below.

Describe core data concepts

Identify considerations for relational data on Azure

Describe considerations for working with non-relational data on Azure

Describe an analytics workload on Azure



Day 4: Task 2

1. Scenario Background

"Paws & Whiskers" is a growing pet shop that aims to improve its business by analysing sales, customer information, and inventory data. Currently, the data is collected manually or stored in spreadsheets. Management is interested in transitioning to Microsoft Azure to

streamline data storage, analysis, and reporting, enabling them to make data-driven decisions.

2. Data Laws and Regulations

Identify and explain the data laws and regulations relevant to handling customer data within the proposal. Ensure you cover the following points:e

- **GDPR Compliance**: Highlight the importance of adhering to the General Data Protection Regulation (GDPR), particularly as it relates to storing and processing customer information.
- **Data Protection Act (DPA) 2018**: Outline how the DPA 2018 may affect the way "Paws & Whiskers" collects and stores data, ensuring compliance with UK laws on data privacy.
- **Other Industry Standards**: Research any additional data protection standards or regulations that may apply to pet shop data, particularly if they involve sensitive or payment information.

3. Azure Service Recommendations

Recommend Microsoft Azure services that would suit the company's data analysis needs and explain why these services are suitable. Your recommendations should include:

- **Data Storage**: Identify suitable storage options, such as **Azure Blob Storage** or **Azure SQL Database**, and discuss the benefits of each for storing large datasets, including inventory, sales transactions, and customer details.
- **Data Analysis Tools**: Recommend tools such as **Azure Machine Learning** for customer behaviour analysis or **Azure Synapse Analytics** for analysing sales trends.
- **Data Integration and Automation**: Explain how services like **Azure Data Factory** could automate data collection and integration processes, improving efficiency.

4. Data Types and Data Modelling

Define the types of data "Paws & Whiskers" will need to work with and describe your approach to data modelling:

- **Data Categories**: Identify key data types, such as customer demographics, transaction history, pet inventory, and product categories.
- **Data Modelling Approach**: Outline how you would structure this data using a relational model or a data warehouse approach, considering factors like tables, entities, relationships, and primary keys.

5. Data Storage Formats and Structures in Azure

Discuss how you would store data within Azure and the formats you would recommend:

- **Data Formats**: Specify recommended formats (e.g., CSV for raw data imports, JSON for structured data, Parquet for analytics) and explain why these formats are suitable for specific data types.
- **Data Security and Encryption**: Include recommendations for securing data using Azure's built-in encryption features and access controls to ensure compliance with data privacy regulations.

6. Additional Considerations

Provide any other considerations that might enhance data handling and efficiency in Azure, such as:

- **Backup and Disaster Recovery**: Outline a backup plan using **Azure Backup** or **Azure Site Recovery** to safeguard against data loss.
- **Data Visualisation**: Discuss potential use of **Power BI** within Azure for creating dashboards that provide management with real-time insights into sales and customer trends.
- **Future Scalability**: Comment on how Azure services can scale as the business grows, accommodating larger datasets and more complex analyses.

Submission Guidelines:

1. **Structure**: Ensure your report is well-organised, with sections for each task (e.g., Data Laws, Azure Services, Data Types, etc.).
2. **Formatting**: Include headings, bullet points where appropriate, and any visuals or diagrams that support your explanations.
3. **References**: Cite any resources or regulations referenced in the report.
4. **Length**: Aim for 1500-2000 words.

To: Management, Paws and Whiskers
Subject: Proposal for modernizing data infrastructure with Microsoft Azure
From: Jonathan Akpofure (Data and Analytics Solutions Consultant)
"Paws and Whiskers" currently relies on manual processes and spreadsheets to store and collect data which greatly limits their ability to extract useful information from sales data, customer data and inventory(product) data. This proposal gives a brief idea of what's expected when shifting to Microsoft Azure, enabling the ability to make data-driven decisions. Microsoft Azure is a cloud platform that automates reporting, streamlines data, and can help with data analysis amongst other things.

In order for this transition to be as smooth as possible, the proposal must strictly adhere to the UK and international data laws. This is important when handling sensitive customer data as to avoid loss of customer trust (reputational damage) and heavy financial penalties.
The General Data Protection Law (GDPR) is critical to the law to consider when handling personal data. Compliance with the GDPR Law requires that you store your data securely and that you ensure customers can exercise their rights; meaning that they have direct control of how the personal data is stored, collected and used.
The core principles of the GDPR Law are the lawfulness, fairness and transparency principle, data minimization, storage limitation, and accountability. Lawfulness, fairness, and transparency relate to having a justifiable reason for wanting to process data. Data minimisation refers to only collecting and storing data that is necessary for the task you want to carry out. It's important to take only what is needed. Storage limitation refers to the fact that customer info can't be kept for an unlimited amount of time. It is important to make it known how long you intend to hold the data and to delete the data you no longer need. Accountability is the last core principle, and it refers to the fact that as a business "Paws and Whiskers" must show compliance. They must take active responsibility to show how they are complying with the GDPR law and proactive measures such as hiring a Data Protection Officer (DPO). It is imperative as accountability helps to build trust and reduce the risk of penalty fines.
The other main act to consider when handling customer data is the Data Protection Act (DPA) 2018. The Data Protection Act allows customers to access their data; rectify any issues they may have with the data and in special cases to completely delete the data which is also known as 'right to be forgotten'. The act also states that the majority of businesses that handle personal data must register with the ICO (Information Commissioners Office) and pay a data protection fee every year. The act requires businesses to make regular staff training on handling data available to everyone. Going from manually collected spreadsheets to Microsoft Azure significantly reduces the risk of breaches occurring. If one does occur, it is compulsory for this to be reported to the ICO within 3 days (72 hours).

Beyond the previously mentioned general privacy laws, "Paws and Whiskers" must consider any additional data protection standards or regulations that may apply to pet shop data, particularly if they involve sensitive or payment information.
One to consider is the PCI DSS (Payment Card Industry Data Security Standard). If card payments are processed through its Azure- hosted system, then it must comply with the PCI DSS to ensure security and storage of credit card data. Another one to consider is the PECR (Privacy and Electronic Communications Regulations). This refers to emails, text messages, and use of cookies on the shop's website. It generally requires consent before sending promotional materials to customers. Others to consider are named as special category of data and animal welfare records which relate to keeping records on medical treatments for the purpose of monitoring health and diseases.
For "Paws and Whiskers" to enjoy a smooth transition to Microsoft Azure, a hybrid-storage approach is best utilized. Azure SQL Database is good when you have structured data such as tables. It should be used for inventory records and the customers' contact details (phone number, email address). A benefit of Azure SQL Database is that it offers automated backups and advanced security. Azure Blob Storage can handle unstructured data like documents and files. It should be used to store images of the product, scanned pet records etc... A benefit of Azure Blob Storage is that it allows different pricing tiers based on how often you need to use the information. It's important to note that Cosmos DB can handle structured and unstructured data together to explore large scale data analytics. This is nice to have when you have big data.
Once you choose the way you want the data to be stored, then you can use tools for data analysis. Azure Synapse Analytics for analyzing sales trends over the years. The benefit of this is that it allows you to query structured data and unstructured data in one sitting. Azure Machine Learning can be used to identify when a customer is due for a purchase of pet food more than once. The benefit of Azure Machine Learning is that it finds trends/patters in the data automatically which could help "Paws and Whiskers" to make personalized adverts targeted to specific types of customer.

The most important step is to automate data collection and integration processes. Azure Data Factory (ADF) can help with this as it can be utilized to collect the data from spreadsheets and to put it into data storage services. There are many benefits of Azure Data Factory, including it can automatically clean data and it's very efficient amongst other benefits.
Moving away from this, it's important to categorize the data and structure it properly.
The shop's data falls into 3 main categories. These are Customer data, Sales data and Inventory (Product) data. Customer data includes things like names and email addresses, customer ID (Primary Key). Sales data includes things like currency, total sales, and sales ID (Primary Key). Inventory data includes things like price, Inventory ID, Inventory type, Inventory name, etc.
A star schema would be best suited because it ensures data is only stored once, and it is easily searchable which could prove beneficial for non-technical people. It's good as it would suit a wide range of people. Customers' table will store unique identity details, inventory table will store prices and brief descriptions of items, and the inventory table connects customers to the product that they bought.
The tables would be arranged in a star format as this structure makes it easier to create reports. This makes it easier to understand, which helps to identify patterns, and in turn allows the business to make data-driven decisions.

I would store the data collected through two different types of data file formats. These data file formats would be CSV and JSON. I would use CSV for any existing spreadsheets the business may have already collected and stored. I would use CSV for this task because almost all apps, databases, systems, and programming languages make it the ideal data file format to transfer data between different systems. I also would use it because they are better suited to handle large datasets compared to something like Excel which might slow down. Also, CSV data files are easy to edit because it contains plain text which makes it easy for anyone to use. I would like JSON for any of the product and customer data because it's ideal for storing difficult product fields/attributes.

Using Microsoft Azure is a good decision for the pet business “Paws and Whiskers” because all the data stored in Microsoft Azure is automatically encrypted using Azure storage service encryption. Access-based control can be used to ensure compliance with data privacy regulations. For example, a regular shop employee may only have access to the different products/inventory but someone with more responsibility like the manager has access to this and has access to see the company’s account, confidential information etc.
Azure Backup is a good service that will provide a safety net and cushion if you happen to lose your data or it gets corrupted. It can be recovered by enabling this service. If an error occurs when importing the data, it can be made to go back to a specific time (e.g. 10 mins ago) so you can restart the process and hopefully the database doesn’t crash.

Data is very useful if non-technical staff can understand it and what it represents. With Power-BI, real-time dashboards can be created to show very recent data. Using tools provided in Power BI, you can filter the data to show specific things you’re interested in, such as sales for dog products in the last 72 hours. This shows it’s a very flexible and powerful tool to mine data.

During periods when more products are expected to be sold, the shop can temporarily increase the power of their computers to deal with the higher sales amount and then once this period is over, they can decrease the power of their computers to reduce the budget.

Course Notes

It is recommended to take notes from the course, use the space below to do so, or use the revision guide shared with the class:

We have included a range of additional links to further resources and information that you may find useful, these can be found within your revision guide.

END OF WORKBOOK

Please check through your work thoroughly before submitting and update the table of contents if required.

Please send your completed work booklet to your trainer.